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HS-OBS	Functional Requirements Specification
DEPARTMENT	DOCUMENT TYPE

VZI R1

Operating Modes Functional Requirements Specification
Lift system

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CHECKED	05/27/2024	S.Tesi	DOORS MODULE NAME	
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Revision History

Rev	Date	Author	Revision Description
00	27/07/2023	G. Tondini	First Issue
01	05/27/2024	G. Guidi	PARAGRAPHS CHANGES: - New Paragraphs: - - Modified Paragraphs: 3.1.1.1.1, 3.1.1.1.3. - Deleted Paragraphs: - REQUIREMENT CHANGES: - New Requirements: 2F_14.Zefiro-Europe.CONCEPT.HR04, 2F_14.Zefiro-Europe.CONCEPT.HR05, 2F_14.Zefiro-Europe.CONCEPT.HR06, 2F_14.Zefiro-Europe.CONCEPT.HR07, 2F_14.Zefiro-Europe.CONCEPT.HR08. - Modified Requirements: 2F_14.Zefiro-Europe.CONCEPT.HR01, 2F_14.Zefiro-Europe.CONCEPT.HR02, 2F_14.Zefiro-Europe.CONCEPT.HR03. - Deleted Requirements: -

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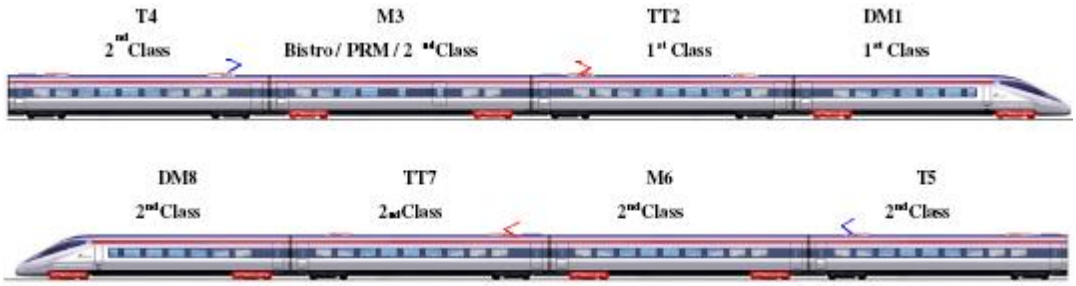
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1 Introduction

This document is a function specification for the control of the Lift system for VZI R1 for customer Trenitalia.

1.1 Train Configuration

Each VZI R1 train consists of eight cars in a fixed formation including a driver's cab at each end:



- DM = End driving motor car with driver's cab
- TT = Intermediate trailer car with transformer
- M = Intermediate driving motor car
- T = Intermediate trailer car

1.2 Definitions and abbreviations

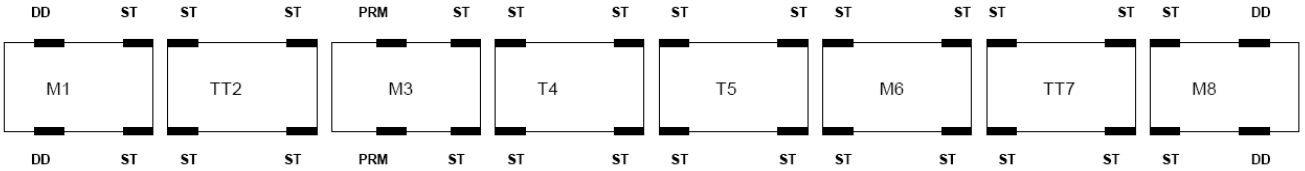
CD	Circuit Diagram
EVENT	Diagnostics that will be stored in the Event database within TCMS SW.
HW HMI	Hardware Human Machine Interface, e.g. buttons, switches and indication lamps.
IDU	Interface Display Unit
TCMS	Train Control and Management System

2 System Overview

2.1 System Configuration

The lift system is made of:

- 2 lift installed in M3 car beside of each PRM doors. These doors can be locally operated only by the train crew and have the exclusive use to allow the access and the egress of wheelchair users though the relevant lift; no other passengers shall be allowed to use these doors.



ST = Passengers standard door

PRM = PRM door for wheel chair users

DD = Drivers door

MAN-MACHINE INTERFACES

Interfaces in driver desk

- IDU (monitoring and status of the lift)
- Crew interfaces
- Emergency device
- isolation device
- Signal lamp

HARDWIRED SIGNALS

1) LIFT parked
2) Lift OK / NOT OK

3 Functional requirements

3.1 LIFT: Status and events

3.1.1.1 Function design

During the service the lift is kept in stowed position until a manual unlocking by train crew is issued.

A lock device maintains the lift in stowed position until a manual unlocking by train crew is issued.

The stowed position of the lift is monitored by a dedicated limit switch.

3.1.1.1.1 Normal condition

The unlock of the lift (not in stowed position) is monitored by a dedicated limit switch.

The limit switch is monitored by DCU and TCMS, the DCU transmits to the TCMS the SW signal “Lift unlock”.

Because use of the lift is intermittent, an automatic periodical check regarding the health of the lift has been foreseen to avoid the non-efficiency of the system.

For this purpose, in condition of lift in stowed position, the TCMS shall activates periodically (for example once a day) and for a predetermined time the dedicated input (110 Vdc on input 5 of P1/Train conn.) which activates the diagnostic function and detects during such activation the status of two outputs of the 2 (P1/Train conn.) and 7 P1/Train conn.) based on the following signal combinations (dataset from DCU to TCMS IPrmLiftStowed):

Lift Connector PIN	2	7	Diagnosti status
Signal	1	0	LIFT OK
Signal	0	1	LIFT NOT OK
Signal	0	0	LIFT NOT OK
Signal	1	1	LIFT NOT OK

If a ‘‘Lift Not OK’’ status is detected, the TCMS shall generate a diagnostic message for maintenance before reaching the condition of non-efficiency of the system.

Because use of the lift is intermittent, the TCMS shall implement a counter of the operations of the PRM lift (Stowed / Not stowed position based on the cycles of S7) by means the status change of the signals 100409 and 100410 (byte 22, bit 0, type U8).

31	IPrmLiftStowed	22	0	U8		IPrmLiftStowed	TC_BI_1DCU1_X_LiftSts	0 = Not stowed 1 = Stowed 2 = No lift installed	Handicap lift is stowed Note: **Only used for PRM door** Default: 0

Once the counter implemented reaches 10000 cycles, the TCMS shall be shown on IDU a maintenance message to advise the maintainer that a check of the lift and the drive belt replacement are required.

3.1.1.1.2 Failure case

a) Stowed lift signal not available

b) Stowed lift signal available also when the lift is not on stowed position.

c) Ethernet connection between DCU and TCMS not active

~~d) Low oil level of lift~~

~~e) Failure of oil level sensor of the lift~~

~~f) Failure of DC/DC converter or other failure of the lift~~

g) Lift NOT OK

3.1.1.1.3 Failure detection

a) The door does not close when requested for lift not in stowed position.

b) Failure not detectable.

c) The condition is detected by TCMs and DCU.

~~d) Output G of lift low and output H of lift high.~~

~~e) Both outputs G and H of lift high~~

~~f) Both outputs G and H of lift low~~

g) The outputs detected can be as follow:

- 7 High (+110 Vdc) and 2 low (0 Vdc)
- 7 Low (0 Vdc) and 2 Low (0 Vdc)
- 7 High (+110 Vdc) and 2 High (+110 Vdc).

3.1.1.2 Requirements

3.1.1.2.1 LIFT status

3.1.1.2.2

ID	2F_14.Zefiro-Europe.TRS.194
Description	PRM lift in "stowed position" and PRM lift operative status (OK/NOT OK)
Safety level	SIL 0

Nr	Description	Type	Derived to	User Interface	SIL
2F_14.Zefiro-Europe.CONCEPT.314	The PRM door shall only be possible to close if the PRM lift is in "stowed position".	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.311	Status of the PRM lift "stowed position" shall be sent to the relevant DCU	Derived Requirement	4S_09.02.09.-.Conventional Train Control - Circuits		SIL 0
2F_14.Zefiro-Europe.CONCEPT.539	DCU shall send to TCMS the current lift status received from PRM lift.	Derived Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.540	The current lift status shall be shown upon request on the IDU	Derived Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR01	The TCMS shall activates periodically (for example once a day) and for a predetermined time the dedicated input (110 Vdc on input 5 of P1/Train conn.)	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR02	The system shall inform the TCMS regarding the status of the PRM lift (OK or NOT OK) by means a combination of the output signals 2 and 7 as detailed in section ‘‘3.1.1.1.1 Normal condition’’.	New Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR03	Based of the feedback received from the system, the TCMS shall be shown the lift status (OK or NOT OK) upon request on the IDU.	New Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0

2F_14.Zefiro-Europe.CONCEPT.HR04	The TCMS shall implement a counter of the operations of the PRM lift (Stowed / Not stowed position based on the cycles of S7) by means the status change of the signals 100409 and 100410.	New Requirement	4S_09.03.05.-.TCMS Software		SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR05	The TCMS shall activate a CBM diagnostic message in order to require a check of the lift and maintenance at next planned maintenance stop of the train.	New Requirement	4S_09.03.05.-.TCMS Software	IDU	SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR06	The Lift shall be equipped parking bypass selector located on door pillar.	New Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR07	When Lift bypass selector is operated, the System shall send to TCMS the activation signal.	New Requirement	4S_12.-.Exterior Door Systems		SIL 0
2F_14.Zefiro-Europe.CONCEPT.HR08	Based on the status of the bypass selector status signal, the TCMS shall visualize the related message on TOD.	New Requirement	4S_12.-.Exterior Door Systems	IDU	SIL 0
~~2F_14.Zefiro-Europe.CONCEPT.768~~	~~The TCMS shall activate periodically, (e.g. once a day), for a predetermined time (e.g. 10 s) and with the lift in stowed position, the diagnostic input of the lift (110 Vdc on input I)~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~		~~SIL 0~~
~~2F_14.Zefiro-Europe.CONCEPT.770~~	~~If the output G of lift is low and output H of lift is high when the diagnostic input is activate a CBM diagnostic message shall be given requiring a visual check of the lift and maintenance at next planned maintenance stop of the train.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~EVENT~~	~~SIL 0~~
~~2F_14.Zefiro-Europe.CONCEPT.772~~	~~If both output G and H of lift are high or low when the diagnostic input is activate a diagnostic message shall be given requiring a corrective maintenance action.~~	~~Derived Requirement~~	~~4S_09.03.05.-.TCMS Software~~	~~EVENT~~	~~SIL 0~~